SQL PORTFOLIO PROJECT

India EV Registration & Charging Infrastructure Analysis



Prepared By :

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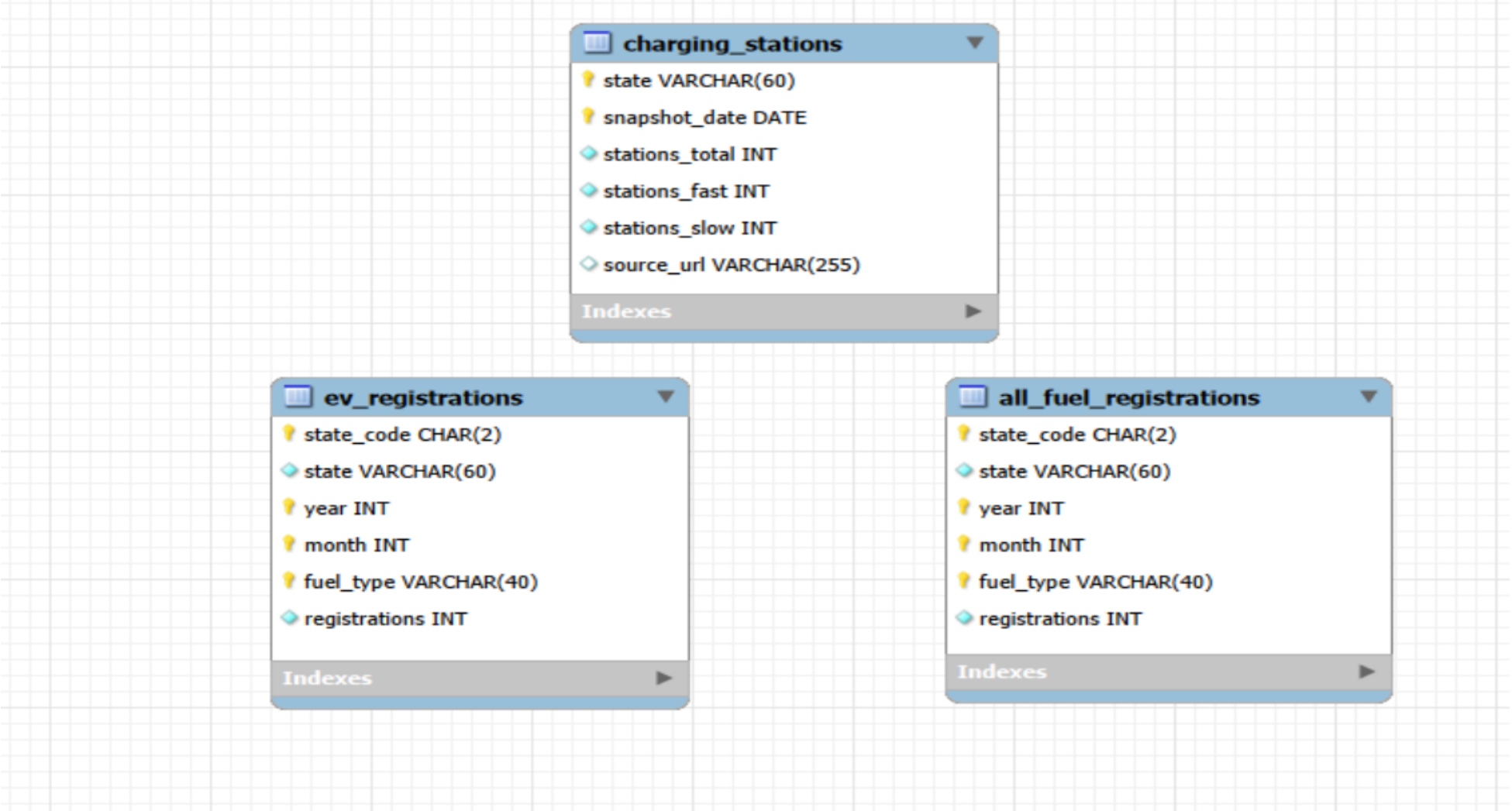
Aspiring Data Analyst

Project overview & business objective

Hi, I'm **Subhajeet Hazra**, an aspiring Data Analyst with hands-on experience in SQL, Excel, Power BI, and Python.
This project analyzes **India's EV registrations and public charging infrastructure** using MySQL. It combines state-level EV registration data from **2021–2025** with a **2025 charging-station snapshot** to explore growth trends, state-wise adoption, EV penetration, and possible infrastructure gaps.

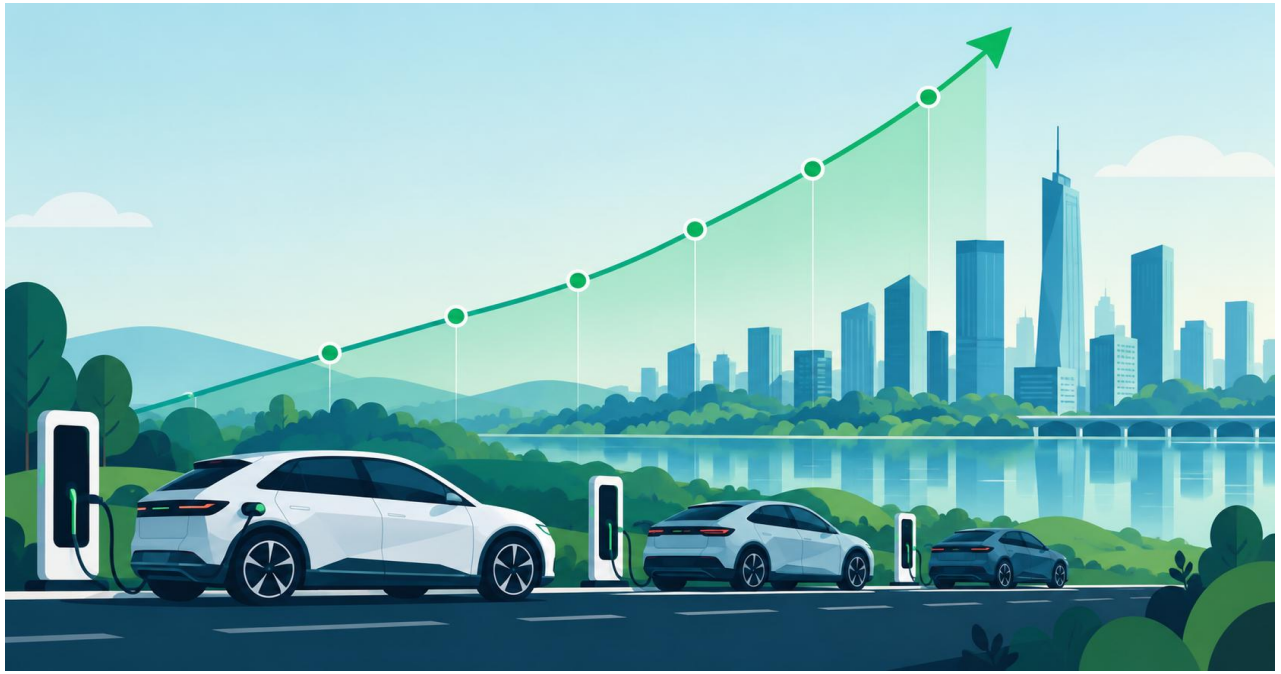


Database Structure



SQL concepts used

- SELECT / WHERE / ORDER BY
- Aggregations
- GROUP BY
- JOINs
- CTEs
- CASE WHEN
- Window Functions
- Percentage & Growth Calculations
- Primary Keys & Constraints



Q1 . How have total EV registrations changed from 2021 to 2025?

SQL query

```
SELECT
    year,
    SUM(registrations) AS total_ev_registrations
FROM ev_registrations
WHERE year BETWEEN 2021 AND 2025
GROUP BY year
ORDER BY year;
```

Result

year	total_ev_registrations
2021	331605
2022	996188
2023	1488643
2024	1881022
2025	2177750

What I found - EV registrations increased every year, rising from about 3.3 lakh in 2021 to 21.8 lakh in 2025. This shows strong and consistent EV adoption over the five-year period.

Q2 . Which 10 states had the highest EV registrations in 2025?

SQL query

```sql
SELECT
    state,
    SUM(registrations) AS total_ev_registrations
FROM ev_registrations
WHERE year = 2025
GROUP BY state
ORDER BY total_ev_registrations DESC
LIMIT 10;
```

Result

state	total_ev_registrations
Uttar Pradesh	403031
Maharashtra	266617
Karnataka	202764
Tamil Nadu	164974
Madhya Pradesh	138963
Bihar	123763
Delhi	112471
Rajasthan	111366
Kerala	104715
West Bengal	98514

What I found - Uttar Pradesh recorded the highest EV registrations in 2025, followed by Maharashtra and Karnataka. This shows that EV registrations are heavily concentrated in a few major states.

Q3 . What was the year-over-year growth in EV registrations from 2021 to 2025?

SQL query

```
WITH yearly_ev AS (
    SELECT year,
        SUM(registrations) AS total_ev_registrations
    FROM ev_registrations
    WHERE year BETWEEN 2021 AND 2025
    GROUP BY year ),
previous_year AS (
    SELECT year,total_ev_registrations,
        LAG(total_ev_registrations) OVER (ORDER BY year) AS previous_year_registrations
    FROM yearly_ev )
SELECT year,total_ev_registrations,
    previous_year_registrations,
    ROUND((total_ev_registrations - previous_year_registrations)
        / previous_year_registrations * 100, 2
    ) AS growth_percentage
FROM previous_year
ORDER BY year;
```

Result

year	total_ev_registrations	previous_year_registrations	growth_percentage
2021	331605	NULL	NULL
2022	996188	331605	200.41
2023	1488643	996188	49.43
2024	1881022	1488643	26.36
2025	2177750	1881022	15.77

What I found - YoY growth peaked at 200.41% in 2022 and gradually slowed to 15.8% in 2025. Registrations are still increasing, but the pace of growth has slowed .

Q4 . Which states recorded the largest increase in EV registrations from 2024 to 2025?

SQL query

```
WITH state_growth AS (
    SELECT state,
        SUM(CASE WHEN year = 2024 THEN registrations ELSE 0 END) AS registrations_2024,
        SUM(CASE WHEN year = 2025 THEN registrations ELSE 0 END) AS registrations_2025
    FROM ev_registrations
    WHERE year IN (2024, 2025)GROUP BY state )
SELECT
    state,registrations_2024,registrations_2025,
    registrations_2025 - registrations_2024 AS absolute_increase,
    ROUND((registrations_2025 - registrations_2024) / registrations_2024 * 100, 2)
AS growth_percentage FROM state_growth ORDER BY absolute_increase DESC;
```

Result

state	registrations_2024	registrations_2025	absolute_increase	growth_percentage
West Bengal	45632	98514	52882	115.89
Madhya Pradesh	100455	138963	38508	38.33
Uttar Pradesh	369286	403031	33745	9.14
Tamil Nadu	131486	164974	33488	25.47
Delhi	80311	112471	32160	40.04
Maharashtra	242598	266617	24019	9.90
Karnataka	179090	202764	23674	13.22
Kerala	83323	104715	21392	25.67
Assam	68131	85002	16871	24.76
Bihar	112857	123763	10906	9.66
Jharkhand	25981	33873	7892	30.38
Jammu and Kas...	8310	11544	3234	38.92
Andhra Pradesh	56695	59868	3173	5.60
Rajasthan	109394	111366	1972	1.80
Sikkim	7	9	2	28.57
Dadra and Nag...	339	300	-39	-11.50
Himachal Pradesh	1746	1536	-210	-12.03
Chandigarh	6199	5143	-1056	-17.04
Chhattisgarh	48844	47393	-1451	-2.97
Goa	11486	9350	-2136	-18.60
Haryana	42705	37333	-5372	-12.58

What I found - West Bengal recorded the largest increase, adding more than 52,000 EV registrations, followed by Madhya Pradesh and Uttar Pradesh. A few states also recorded a decline compared with 2024 like Haryana,Goa.

Q5. What percentage of total vehicle registrations were EVs in each state in 2025?

SQL query

```sql
WITH ev_2025 AS (
    SELECT state,
        SUM(registrations) AS ev_registrations
    FROM ev_registrations WHERE year = 2025
    GROUP BY state),
total_2025 AS (
    SELECT
        state, SUM(registrations) AS total_vehicle_registrations
    FROM all_fuel_registrations
    WHERE year = 2025
    GROUP BY state)
SELECT
    e.state,t.total_vehicle_registrations,
    e.ev_registrations,
    ROUND(e.ev_registrations / t.total_vehicle_registrations * 100, 2)
AS ev_penetration_percentage FROM ev_2025 e
JOIN total_2025 t ON e.state = t.state
ORDER BY ev_penetration_percentage DESC;
```

Result

state	total_vehicle_registrations	ev_registrations	ev_penetration_percentage
Tripura	55569	10198	18.35
Assam	595217	85002	14.28
Delhi	817127	112471	13.76
Kerala	865056	104715	12.10
Goa	86297	9350	10.83
Karnataka	1907499	202764	10.63
Chandigarh	50458	5143	10.19
Uttar Pradesh	4076646	403031	9.89
Madhya Pradesh	1687799	138963	8.23
Bihar	1505941	123763	8.22
Maharashtra	3289485	266617	8.11
Tamil Nadu	2108521	164974	7.82
West Bengal	1277669	98514	7.71
Rajasthan	1626151	111366	6.85
Andaman and N...	9665	166	1.72
Himachal Pradesh	129960	1536	1.18
Dadra and Nag...	27169	300	1.10
Arunachal Prad...	41056	103	0.25
Nagaland	27058	23	0.09
Sikkim	10416	9	0.09

What I found - Tripura had the highest EV penetration at 18.35%, followed by Assam and Delhi. This shows that the states with the most EVs are not necessarily the states where EVs make up the largest share of vehicle registrations.

Q6. Which states had the most public EV charging stations, and what share were fast chargers?

SQL query

```sql
SELECT
    state,
    stations_total,
    stations_fast,
    stations_slow,
    ROUND(stations_fast / stations_total * 100, 2)
    AS fast_charger_percentage
FROM charging_stations
ORDER BY stations_total DESC;
```

Result

state	stations_total	stations_fast	stations_slow	fast_charger_percentage
Karnataka	6096	751	5345	12.32
Maharashtra	4166	1205	2961	28.92
Uttar Pradesh	2316	540	1776	23.32
Delhi	1957	399	1558	20.39
Tamil Nadu	1780	926	854	52.02
Rajasthan	1515	523	992	34.52
Kerala	1389	553	836	39.81
Gujarat	1205	578	627	47.97
Madhya Pradesh	1146	353	793	30.80
Telangana	1062	548	514	51.60
Haryana	919	384	535	41.78
West Bengal	891	321	570	36.03
Andhra Pradesh	791	424	367	53.60
Punjab	698	225	473	32.23
Chandigarh	13	11	2	84.62
Mizoram	13	0	13	0.00
Sikkim	12	3	9	25.00
Andaman and N...	4	1	3	25.00
Ladakh	1	0	1	0.00
Lakshadweep	1	0	1	0.00

What I found - Karnataka had the most public charging stations, followed by Maharashtra and Uttar Pradesh. However, the share of fast chargers varied significantly between states.

Q7. Which states had the highest number of 2025 EV registrations per public charging station?

SQL query

```sql
WITH ev_2025 AS (
    SELECT state,
        SUM(registrations) AS ev_registrations_2025
    FROM ev_registrations
    WHERE year = 2025
    GROUP BY state )
SELECT
    e.state,e.ev_registrations_2025,
    c.stations_total,
    ROUND(e.ev_registrations_2025 / c.stations_total, 2) AS registrations_per_station
FROM ev_2025 e
JOIN charging_stations c
    ON e.state = c.state
ORDER BY registrations_per_station DESC;
```

Result

state	ev_registrations_2025	stations_total	registrations_per_statior
Chandigarh	5143	13	395.62
Assam	85002	345	246.38
Bihar	123763	518	238.92
Tripura	10198	55	185.42
Uttar Pradesh	403031	2316	174.02
Chhattisgarh	47393	347	136.58
Madhya Pradesh	138963	1146	121.26
West Bengal	98514	891	110.57
Puducherry	4693	46	102.02
Jharkhand	33873	353	95.96
Tamil Nadu	164974	1780	92.68
Uttarakhand	20053	221	90.74
Andhra Pradesh	59868	791	75.69
Kerala	104715	1389	75.39
Dadra and Nag...	300	14	21.43
Manipur	1201	58	20.71
Meghalaya	930	62	15.00
Himachal Pradesh	1536	133	11.55
Arunachal Prad...	103	45	2.29
Sikkim	9	12	0.75
Nagaland	23	37	0.62

What I found - Chandigarh, Assam and Bihar recorded the highest EV registrations per public charging station, indicating relatively higher pressure on charging infrastructure.

Q8. Among the top 10 EV states in 2025, which had the lowest share of fast chargers?

SQL query

```sql
WITH top_states AS (
    SELECT state,
    SUM(registrations) AS ev_registrations_2025
    FROM ev_registrations WHERE year = 2025
    GROUP BY state ORDER BY ev_registrations_2025 DESC
    LIMIT 10)
SELECT t.state, t.ev_registrations_2025,
c.stations_total,c.stations_fast,
ROUND(c.stations_fast / c.stations_total * 100, 2)
AS fast_charger_percentage
FROM top_states t
JOIN charging_stations c
ON t.state = c.state
ORDER BY fast_charger_percentage;
```

Result

state	ev_registrations_2025	stations_total	stations_fast	fast_charger_percentage
Karnataka	202764	6096	751	12.32
Delhi	112471	1957	399	20.39
Uttar Pradesh	403031	2316	540	23.32
Maharashtra	266617	4166	1205	28.92
Madhya Pradesh	138963	1146	353	30.80
Bihar	123763	518	172	33.20
Rajasthan	111366	1515	523	34.52
West Bengal	98514	891	321	36.03
Kerala	104715	1389	553	39.81
Tamil Nadu	164974	1780	926	52.02

What I found - Among the leading EV states, Karnataka had the lowest fast-charger share at 12.32%, while Tamil Nadu had the highest at about 52%. This shows that having more charging stations does not necessarily mean having better fast-charging availability.

Key insights

- EV registrations increased strongly from 2021 to 2025.
- Uttar Pradesh led 2025 EV registrations.
- West Bengal recorded the largest absolute growth from 2024 to 2025.
- EV penetration and total EV volume did not always move together.
- Charging-station availability and fast-charger share varied significantly across states.
- Some states showed high registrations relative to charging-station availability.